FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – X
PAPER –2
TEST DATE: 15-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.
  Section – A (01 – 04, 18 – 21, 35 – 38): *This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*
  Section – A (05 –07, 22 – 24, 39 – 41): *This section contains **NINE (09)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).*
  Section – B (08 – 13, 25 – 30, 42 – 47): *This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*
  Section – C (14 –17, 31 – 34, 48 – 51): *This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Physics

PART – I

SECTION – A
(One Options Correct Type)

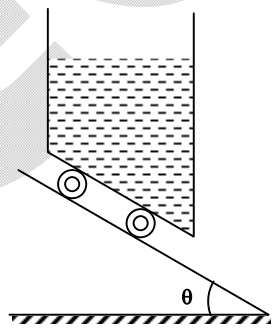
*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

1. Two concentric metal spheres of radii 'a' and 'b' (b > a) are separated by a medium that has dielectric constant ε_r and conductivity 'σ'. At time t = 0 an electric charge 'q_0' is suddenly placed on the inner sphere. What is the total current flowing through the medium at time 't'.

(A) $\frac{\sigma q_0}{\varepsilon_r \varepsilon_0} e^{-\frac{\sigma}{\varepsilon_r \varepsilon_0} t}$ (B) $\frac{\sigma q_0}{2\varepsilon_r \varepsilon_0} e^{-\frac{\sigma}{\varepsilon_r \varepsilon_0} t}$

(C) $\frac{\sigma q_0}{4\varepsilon_r \varepsilon_0} e^{-\frac{\sigma}{\varepsilon_r \varepsilon_0} t}$ (D) $\frac{2\sigma q_0}{\varepsilon_r \varepsilon_0} e^{-\frac{\sigma}{\varepsilon_r \varepsilon_0} t}$

2. A body floats in completely immersed condition in water as shown in the figure. As the whole system is allowed to slide down freely along the inclined surface, the magnitude of buoyant force
(A) remains unchanged
(B) increases
(C) decreases
(D) become zero



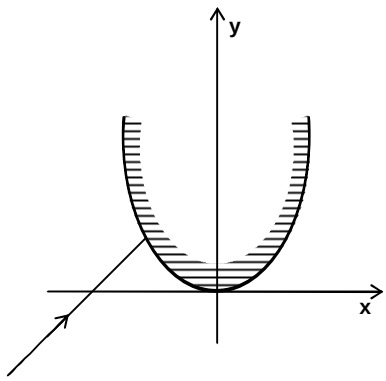
3. You are given parabolic mirror whose inner surface is silvered so that its outer surface behaves like a mirror. The equation of the curve formed by its intersection with xy plane is given by $y = \frac{x^2}{4}$. A ray travelling in xy plane along line y = x +3, hits the mirror in second quadrant and gets reflected. The unit vector in the direction of reflected ray is :

(A) $\frac{1}{\sqrt{2}}(-\hat{i} - \hat{j})$

(B) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$

(C) $\frac{1}{\sqrt{2}}(-\hat{i} + \hat{j})$

(D) none of these



4. $^{239}_{94}Pu$ is undergoing α-decay according to the equation

$$^{239}_{94}Pu \rightarrow ^{235}_{92}U + ^{4}_{2}He$$

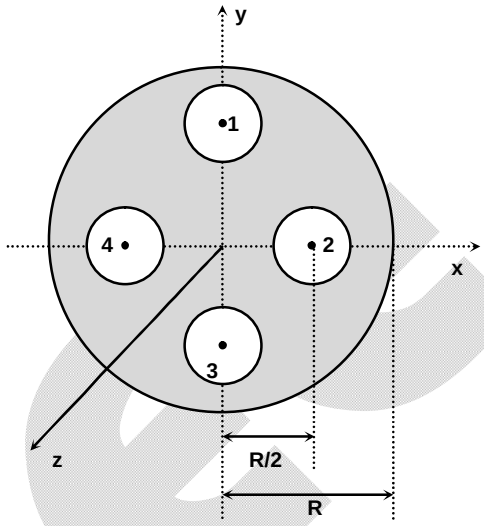
The process is accompanied by an emission of γ-rays of wavelength 0.125 Å. Find the Kinetic energy of the emitted α-particle. Given mass of $^{239}_{94}Pu$ = 239.052158 amu, mass of $^{235}_{92}Pu$ = 235.043925 amu, mass of $^{4}_{2}He$ = 4.002603 amu and 1 amu = 931 MeV.

(A) 2.06 Mev (B) 8.03 Mev
(C) 5.06 Mev (D) 12.06 Mev

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. Four identical sphere of radius $\frac{R}{4}$ are taken out from a uniform sphere of mass M and radius R as shown in the figure. The density of sphere is uniform. Choose the correct option(s).



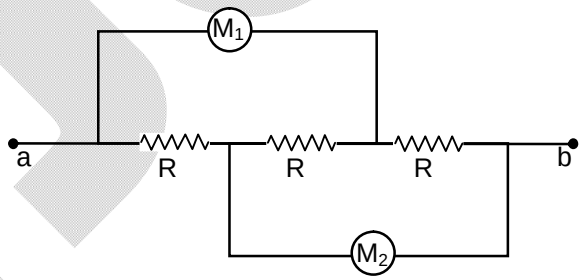
(A) Gravitational field at the origin is zero.

(B) Gravitational potential at the centre of the spherical cavity is $\frac{GM}{64R}(81-2\sqrt{2})$.

(C) The work done by gravitational field in moving a point mass from centre of third cavity to centre of second cavity is zero.

(D) Gravitational potential at the centre of second cavity is $-\frac{GM}{64R}(81-2\sqrt{2})$.

6. In the given circuit the potential difference between a and b is 300V. The resistors are of 100Ω each.



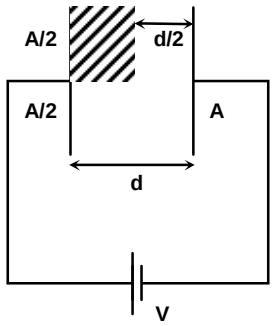
(A) If the upper meter M_1 is ideal voltmeter and lower is ideal ammeter M_2, reading of voltmeter is 300V.

(B) If both meters are ideal voltmeters, they show 200V each.

(C) If both meters are ideal ammeters, they read 6A each.

(D) If the upper meter M_1 is ideal voltmeter and lower is ideal ammeter M_2, reading of ammeter is 3A .

7. $\left(\frac{1}{4}\right)^{th}$ of the region between parallel plate capacitor is filled with dielectric (k = 2) as shown. If σ_1 and σ_2 are the charge density on upper half and lower half plate of capacitor, then:



(A) $\sigma_1 = \frac{4\varepsilon_0 V}{3d}$

(B) $\sigma_2 = \frac{\varepsilon_0 V}{d}$

(C) $\sigma_1 = \sigma_2$

(D) $\sigma_1 > \sigma_2$

SECTION – B

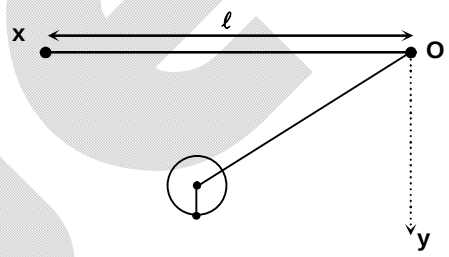
(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

8. Cylinder, solid sphere are placed on top of rough incline plane, when both of them are released at same instant from same point, sphere can just roll without slipping. Find the ratio of their time to reach ground is $\frac{t_{cylinder}}{t_{sphere}}$.

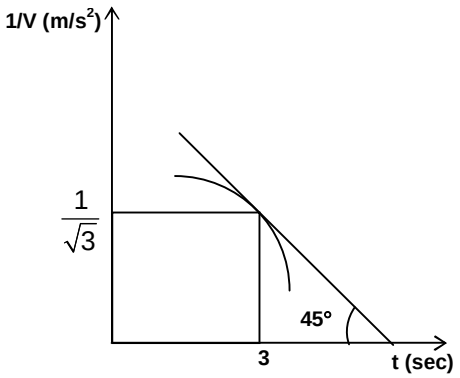
9. A balloon moves up vertically such that if a stone is thrown from it with a horizontal velocity v_0 relative to it the stone always hits the ground at a fixed point $\frac{2v_0^2}{g}$ horizontally away from it. The height of the balloon as a function of time is $\frac{kv_0^2}{g}\left[1-e^{-gt/2v_0}\right]$. Find the value of k.

10. A simple pendulum of length ℓ is suspended by a nail on a vertical wall at point O. Another nail is to be fixed on the wall such that if the pendulum is released from its initial horizontal position, it just completes a vertical circle round that nail as shown. The locus of the point where the second nail can be fixed is $K(x^2+y^2)=(3\ell-2y)^2$. Find the value of K.



11. In a park there are three concentric circular running tracks. Radius of 2nd track is double the radius of first track and the radius of 3rd track is triple the radius of first track. Three runners are running on these tracks with constant speeds. When the runner in the first track completes one round, the runner in 2nd has completed half round and the runner in third track has completed quarter round. If the accelerations of the runners are in ratio $\alpha : \beta : \gamma$ where α, β and γ are least integers, then find the value of $\left(\frac{\alpha+\beta+\gamma}{3}\right)$.

12. The graph shown the variation of $\frac{1}{V}$ (where V is the velocity of the particle) with respect to time. Find the value of acceleration at t = 3 sec in m/s^2.



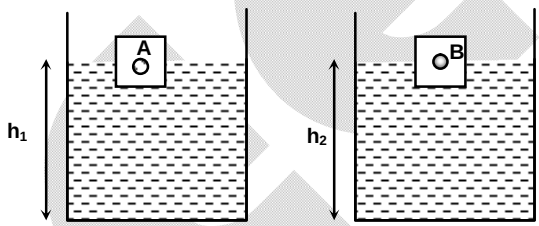
13. A calorimeter contain ice at 270 K. Heat capacity of the calorimeter is $\frac{n}{2}\times 10^2$ cal/°C, where n is single digit number Q_1 = 500 cal are required to heat is together with its contents from 270 K to 272 K and Q_2 = 16600 cal from 272 K to 274 K. Find n. (Specific heat capacity of water is 1 cal/g°C. Specific heat capacity of ice = 0.5 cal/g°C. Latent heat of fusion = 80 cal/g)

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 14 and 15

Two ice cubes of side 10cm , having cavity of volume $20cm^3$ at centre of cube but filled with different materials A and B respectively. The specific gravity of material A is 4.9 and specific gravity of material B is 1.9 . Now these cubes are placed in two different vessels of same base area as shown in figure. The water level before putting blocks in vessels are same. Assume that ice melts uniformly from all sides and with same constant rate in both the vessels. (specific gravity of ice = 0.9)

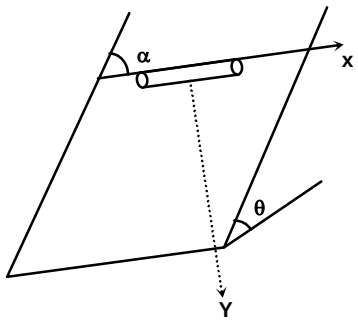


14. The ratio of initial submerged volumes of the blocks containing A and B is $\frac{49}{92k}$. Find the value of k.

15. The depth of cube (having material A) sinked in water initially is k cm. Find the value of k.

Paragraph for Question Nos. 16 and 17

A cylinder of mass m is kept on a inclined plane having angle of inclination 30°. Axis of cylinder makes an angle 30° with line along greatest slope. Assuming that cylinder rolls without slipping:



16. Acceleration of cylinder is $\frac{g}{60k}$. The value of k is…..

17. Friction force acting on cylinder is $\frac{\sqrt{7}mg}{30P}$. The value of P is ….

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

18. pH of a saturated solution of silver salt of monobasic acid HA is found to be 9. Find out solubility product of sparingly soluble salt AgA(s) in $mole^2/L^2$. [Given: K_a of $HA = 10^{-10}$]
(A) 1.1×10^{-11} (B) 1.1×10^{-8}
(C) 10^{-12} (D) 1.1×10^{-5}

19. For the reaction, $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
$\Delta H^o = 57.5$ kJ and $\Delta S^o = 174\ JK^{-1}$.
Determine the temperature at which on equilibrium mixture with a total pressure 20 atm contains equal moles of NO_2 and N_2O_4.
(Use $2.303R = 19\ JK^{-1}$)
(A) 330.5 K (B) 298 K
(C) 371 K (D) 401.5 K

20. Although NF_3 and NCl_3 are covalent, NCl_3 undergoes hydrolysis whereas NF_3 does not because:
(A) NF_3 is more stable than NCl_3
(B) dipole moment of NF_3 is less than NCl_3
(C) electronegativity of F is more than Cl
(D) Cl can expand its octet by using d-orbitals

21. $(NH_4)_2Cr_2O_7$ on heating liberates a gas. The same gas will be obtained by
(A) heating NH_4NO_2 (B) heating NH_4NO_3 at 250^oC
(C) treating Mg_3N_2 with water (D) treating H_2O_2 and $NaNO_2$

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

22. Which of the following have square planar geometry?
(A) $[Ni(DMG)_2]$ (B) $[NiCl_4]^{2-}$
(C) $[Ni(CN)_4]^{2-}$ (D) $[PtCl_4]^{2-}$

23. Which of the following statements is/are correct about polymers?
(A) Neoprene is a polymer of isoprene
(B) Glyptal is a condensation polymer of ethylene glycol and phthalic acid.
(C) Terylene is a condensation polymer of ethylene glycol and terephthalic acid.
(D) Nylon-6,6 is a condensation polymer of hexamethylenediamine and adipic acid.

24. Which of the following group is/are ortho-para directing and deactivating for electrophilic aromatic substitution reaction?

(A) $-CH=CH-NO_2$ (B) $-Cl$

(C) $-SO_3H$ (D) $-CN$

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

25. How many of the following ethers cannot be prepared by Williamson's synthesis?

(a) $H_3C—O—CH_2—CH_3$

(b) $H_5C_6—O—CH_3$

(c) $H_5C_6—O—C_6H_5$

(d) $(H_3C)_3C—O—CH_3$

(e) $C_6H_5CH_2—O—CH_3$

(f) $C_2H_5—O—C_2H_5$

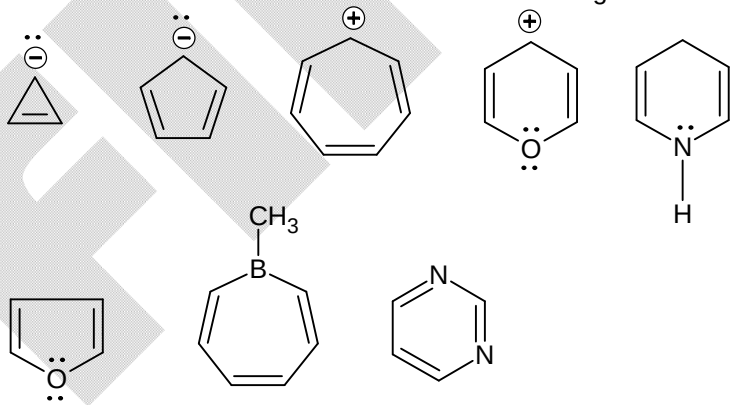
(g) $C_6H_5CH_2—O—C_6H_5$

(h) $(H_3C)_3C—O—C(CH_3)_3$

26. How many ores are sulphide in the following?

(i) Azurite
(ii) Chalcocite
(iii) Iron Pyrites
(iv) Limonite
(v) Malachite
(vi) Siderite
(vii) Pyrolusite
(viii) Argentite
(ix) Cinnabar

27. The total number of aromatic molecules/ions among the following is:



28. When formaldehyde is treated with ammonia, a crystalline compound urotropine is formed. If the number of carbon atoms and nitrogen atoms in the compound are x and y respectively, then the value of $\frac{x}{y}\times 10$ is:

29. A hydrogen like species with atomic number 'Z' is in higher excited state 'n' and emits photons of energy 25.7 eV and 8.7 eV when making a transiton to 1st and 2nd excited state respectively. Calculate the value of 'n'.

30. In faintly alkaline or neutral solution, MnO_4^- oxidizes thiosulphate ion to a sulphur containing ion (P). For this reaction, the ratio of rate of change of $\left[MnO_4^-\right]$ to the rate of change of [P] is 'x'. Calculate the value of 100x.

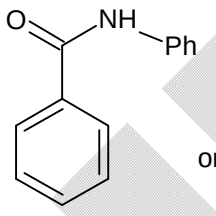
SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

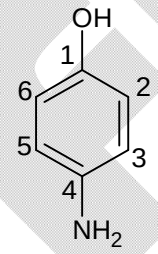
Paragraph for Question Nos. 31 and 32

When mono substituted benzene derivative, C_6H_5Y, undergoes further electrophilic substation, e.g. nitration, the incoming substituent may be incorporated at the o^-,m^- or $p-$position and the overall rate at which substitution takes place may be faster or slower than with benzene itself. It has been observed that substitution occurs so as to yield either predominatly the m-isomer or a mixture of o^- & p^- isomers, in the former case the overall rate of attack is always slower than on beonzene, in the latter case the overall rate of attack is usually faster than on benzene itself. This is due to electronic effect that Y can exert.

When a mono substituted benzene derivative,
Substituent Y is thus classified as o^-/p^- directing; if the substituent induces faster overall attack than on benzene itself then the substituent is said to be activating, if slower then deactivating.
Based on this, answer the following questions:

31. 

on mononitration gives the major product. Find double bond equivalence of major product.

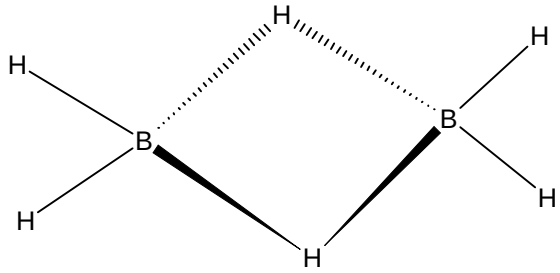
32. 

$+ PhN_2^{\oplus} \xrightarrow{pH=9-10} X(\text{Major product})$. X will be

For major product formation, incoming electrophile will come at position number:

Paragraph for Question Nos. 33 and 34

In diborane, there are four terminal H and two bridging H. The four terminal B – H bonds are regular two centre-two electron bonds while the two bridge (B – H – B) bonds are different and can be described in terms of three centre-two electron bonds.



Based on this answer the following questions

33. Maximum number of atoms in B_2H_6 that can be cleaved by a plane:

34. $N(CH_3)_3$, NH_3, CO, THF, CH_3-NH_2, Et_2O.

How many molecules from the above list can show symmetrical cleavage with B_2H_6.

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

35. If $f : R \to R$, $f(x) = \frac{x^2 + bx + 1}{x^2 + 2x + b}$, (b > 1) and f(x), $\frac{1}{f(x)}$ have the same bounded set as their range, the value of b is

(A) $2\sqrt{3} - 2$ (B) $2\sqrt{3} + 2$

(C) $2\sqrt{2} - 2$ (D) $2\sqrt{2} + 2$

36. In a planar rectangular coordinate system XOY, the area enclosed by the graph of function f(x) = a sin ax + cos ax (a > 0) defined on an interval with the least positive period and by the graph of function $g(x) = \sqrt{a^2 + 1}$ is

(A) $\frac{\pi}{a}\sqrt{a^2 - 1}$ (B) $\frac{2\pi}{a}\sqrt{a^2 + 1}$

(C) $\frac{\pi}{2a}\sqrt{a^2 + 1}$ (D) $\frac{4\pi}{a}\sqrt{a^2 + 1}$

37. Let $f : [a, b] \to (0, \infty)$, where a, b > 0 and b – a < 1 be a continuous function such that $\int_a^b f(x)dx = 1$, then the value of $\left(\int_a^b \sqrt[2n+1]{f(x)}\,dx\right)\left(\int_a^b \sqrt[2n+3]{f(x)}\,dx\right)\left(\int_a^b \sqrt[2n+5]{f(x)}\right)$, $n \in N$ can be

(A) $\frac{1}{2}$ (B) $\frac{4}{3}$

(C) 2 (D) $\frac{3}{2}$

38. Given that the inequality $|2^x - a| < |5 - 2^x|$ holds for all $x \in [1, 2]$, then the range of real number a

(A) (3, 6) (B) (–3, 5)

(C) (3, 5) (D) (5, 9)

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

39. Let $\vec{u} = a\hat{i} + b\hat{j}$ and $\vec{v} = c\hat{i} + d\hat{j}$ be position vector of two points P and Q respectively and A be the matrix representing $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $|A| \neq 0$ satisfying det(A + det(A)·Adj(A)) = 0, then

(A) if $\vec{u}'$ and $\vec{v}'$ be the vectors representing image of P w.r.t. x-axis and image of Q w.r.t y = x line respectively, then $\vec{u}' \cdot \vec{v}' = 1$

(B) $d^2 = 1 - bc$

(C) det(A – (det(A))·Adj(A)) = 4

(D) $\vec{u} \cdot \vec{v} = 1$

40. Let x_1 and x_2, $|x_1| > |x_2|$ be two integral values of x for which values of y be also integer satisfying relation $x^2y + xy^2 = 5 + x^3 + y^3$ and f(x) be a polynomial satisfying
(i) $\prod_{r=1}^{xy} f(r) < 0$, $\prod_{r=1}^{|x_1|} f(2r) < 0$, $\prod_{r=1}^{|x_2|} f(3r) < 0$; (ii) f(xy) > 0 and (iii) f is bijective in (n, n + 1) $n \in I$, then
(A) there exist only two values of the ordered pair (x, y)
(B) there exist minimum three real roots of the equation f′(x) = 0 in R
(C) the value of sum of integral values of the roots of f(x) = 0 when f(x) = 0 has maximum number of real roots in [1, 6] is 15
(D) number of real roots in (2, 4) in two

41. Given that the divisors of $N = 3^{\alpha}\cdot 5^{\beta}\cdot 7^{\gamma}$ are of the form $4\lambda + 1$, $\lambda \geq 0$, then
(A) $\alpha + \beta + \gamma$ may be even or odd
(B) β can be any positive integer
(C) $\alpha + \gamma$ is always even
(D) if α is even, then γ is odd

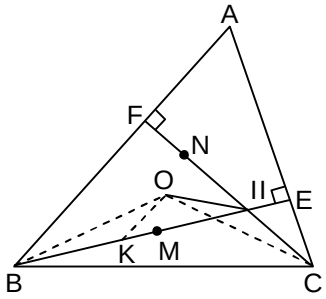
SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

42. If $\frac{\sin 5}{\sin 4} - \frac{\sin 1}{\sin 2} - \frac{\sin 10}{2\cos 1\cdot \sin 8} = \frac{\sin\alpha}{\sin 8}$ and $\beta = \left[\left|\sqrt{2}\sin\left(\frac{\pi\alpha_{min}}{4}\right)^{\circ}\right|\right]$ (where [.] represents greatest integer function) and α_{min} is smallest positive value of α, then number of solution of $\beta\cdot\sin^{2\beta} x = \frac{|x|}{7}$ is

43. As shown in the diagram, in ΔABC, $\angle A = 60°$, AB > AC, point O is a circumcenter and H is the intersection point of two altitudes BE and CF. Points M and N are on the line segments BH and HF respectively, and satisfy BM = CN. Then the value of $\left[\frac{MH + NH}{OH}\right]$ is equal to
(Where [.] represents greatest integer function)



44. If $x, y, z \in \left(0, \frac{\pi}{2}\right)$ and $\sqrt[3]{\left(\sec^{4n} x + \text{cosec}^{4n} x\right)\left(\sec^{6n} y + \text{cosec}^{6n} y\right)\left(\sec^{2n} z + \text{cosec}^{2n} z\right)} = 2\cdot 2^{3^r}$, k, n, m, r are equal natural numbers, then the value of $\sin^{2k} 2x + \sin^{3n} 2y + \sin^{4m} 2z$ is equal to

45. If all the roots of the equation $x^8 - 4x^7 + b_6x^6 + b_5x^5 + b_4x^4 + b_3x^3 + b_2x^2 + b_1x + \frac{1}{2^8} = 0$ are positive, then the number of common tangents to the curve $\frac{x^2}{b_6} + \frac{y^2}{b_5} = 7$ and $x^2 + y^2 = 256b_2^2$ is

46. If f is a real valued function satisfying the relation $1 + f(x + y) \leq f(x) + f(y) \leq \sin x + \sin y + 2$, $\forall$ x, y, then f is a periodic function with fundamental period $k\pi$, then k is equal to

47. If a, b, c, d, e, f, g, h, k $\in R^+$ and a + b + c = d + e + f = g + h + k = 8, then maximum value of mean of
$\sqrt[2]{ad^3g^4}$, $\sqrt[3]{be^3w^4}$, $\sqrt[8]{cf^5k^4}$ is x, then [x] is (where [.] represents greatest integer function)

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 48 and 49

Consider a sequence of real numbers $(a_n)_{n\geq 1}$ defined by: $a_n = \begin{cases} 1 & \text{if } n \leq k, \\ \dfrac{(n+1)^k - n^k}{\binom{n}{k-1}} & \text{if } n > k;\ k \in N \end{cases}$

48. Let $\lim\limits_{n\to\infty} a_n$ is equal to f(k), then f(4) is

49. If $b_n = 1 + \sum\limits_{k=1}^{n} k \cdot \lim\limits_{n\to\infty} a_n$, and $\lim\limits_{n\to\infty}\left(\dfrac{b_n^2}{b_{n-1}b_{n+1}}\right)^n$ = L, then $\left[\dfrac{1}{L}\right]$ is
(where [.] represents greatest integer function)

Paragraph for Question Nos. 50 and 51

Let $f(x) = x^3 - 6x^2 + 14x - 12$ and $g(x) = x^2 - ax + b$ where a, b $\in$ N have two common roots

50. If α, β are the roots of g(x) = 0, then $\left(\dfrac{1}{4-\alpha}\right)^2 + \left(\dfrac{1}{4-\beta}\right)^2$ = k, then 9k is equal to

51. Number of integral values of x, such that $8 < g(x) \leq 18$ is